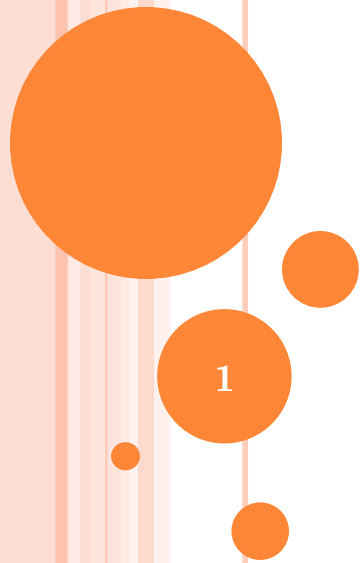




## UNIT 5

# Biomass energy

# SYLLABUS

5.1 Types of Biomass Energy Sources

5.2 Energy content in biomass of different types

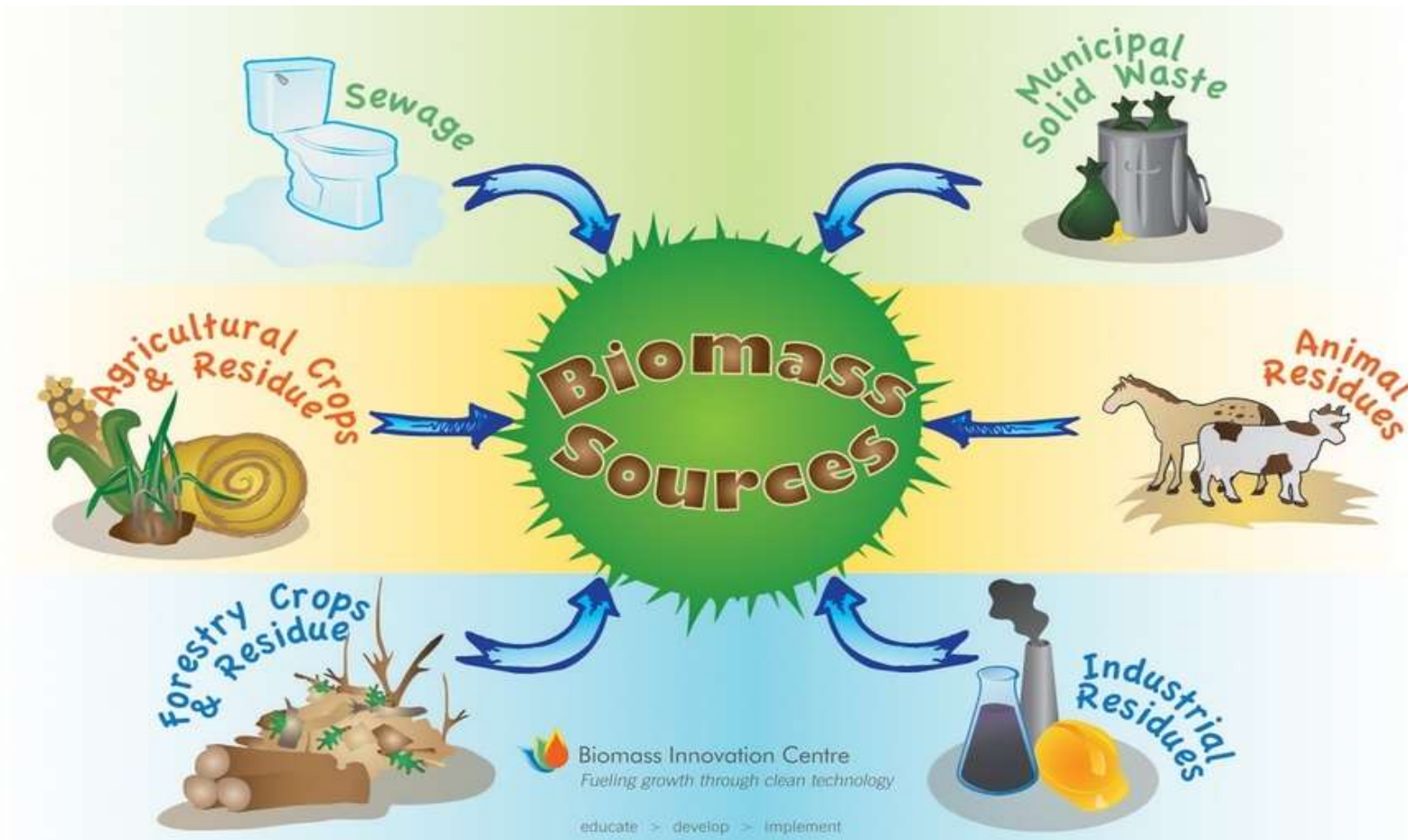
5.3 Types of Biomass conversion processes

5.4 Biogas production

# WHAT IS BIOMASS?

- An organic matter and non conventional sources of Energy
- Derived from living and recently living organisms and organic matters like
  - Plants, Leaves,
  - Sewage
  - Dropping of birds
  - Municipal waste
  - Industrial waste

## 5.1 TYPES OF BIOMASS ENERGY SOURCES



## 3 FORMS OF BIOMASS

- 1) Solid Biomass
- 2) Liquid Biomass
- 3) Gaseous Biomass

### 1) Solid Biomass

- Leaves of plants
- Stems of tree
- Cuttings of tree and plants
- Dried sugar canes
- Forest products
- Aquatic plants such as moss, lilies, marine alga

## 3 FORMS OF BIOMASS

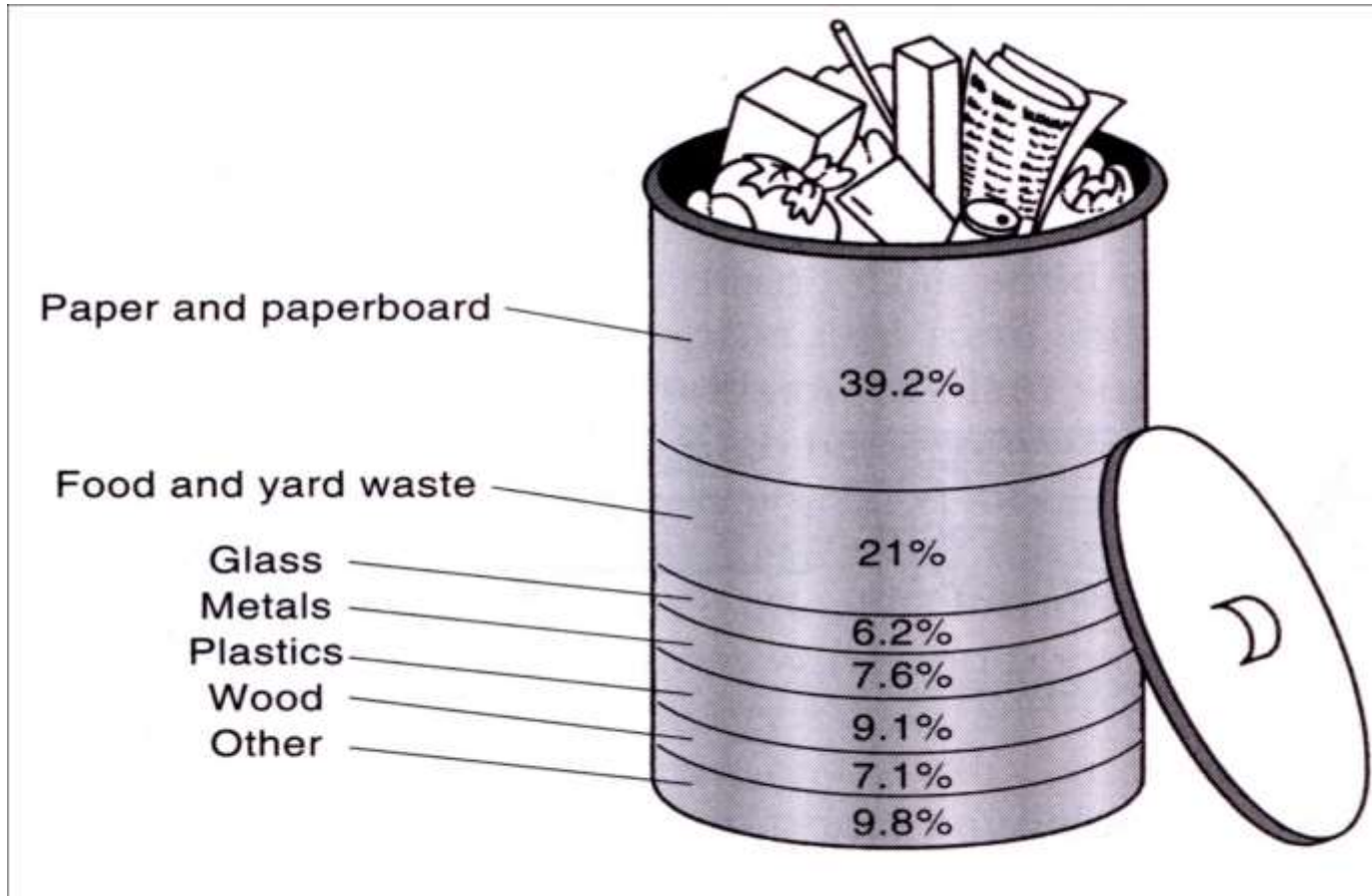
### 2) Liquid Biomass

- Obtain from chemical process carried out on the solid biomass such cow dung and other animal waste.
- They form methyl alcohol, methanol & ethanol from fermentation of biomass
- Used as fuel.

### 3) Gaseous Biomass

- It produced from undergoing anaerobic digestion in biogas plants.
- It is called Methane and Ethane.
- Used as fuels and burn like petroleum gas i.e. LPG

# COMPOSITION OF URBAN GARBAGE



# ADVANTAGES OF BIOMASS ENERGY

- Source of Renewable Energy
- No Harmful Emissions
- Keep environment Clean
- Reduce Dependency on Fossil Fuels
- Reduce Landfills
- Can be Used to run engines and generator.
- Available everywhere & Do not need transportation
- Also grown in nearby seas and lakes
- Provide clean and healthy Energy

# DISADVANTAGES OF BIOMASS ENERGY

- Contain 50 to 90% water and it is heavy
- Direct biomass produce smells & smokes
- Plants need lot of care & maintenance
- Expensive
- Inefficient as Compared to Fossil Fuels
- Harmful to Environment
- Require More Land
- Can not use in Urban areas

## 5.2 ENERGY CONTENT IN BIOMASS OF DIFFERENT TYPES

| Biomass             | Energy content |
|---------------------|----------------|
| Sugar Cane stalks   | 9.6 kj/kg      |
| Cow & Buffalo gobar | 15 kj/kg       |
| Dried Leaves        | 10-16 kj/kg    |
| Charcoal            | 30 kj/kg       |
| Ethanol             | 20 kj/kg       |
| Methanol            | 25 kj/kg       |
| Butadin             | 36 kj/kg       |
| Fat                 | 37 kj/kg       |
| Biodiesel           | 38 kj/kg       |
| Edible oil          | 40 kj/kg       |

# ENERGY PLANTS

- Plants which are not useful in any other way except to be used as fuel are called energy plants,

## Classification

- Rapid growing plants
- Oil yielding plants

# RAPID GROWING PLANTS

- Growing along river, lake and seas
- Grow in water
- Used in pharmaceutical and paper industry.
- See woods, water hyacinth, algae, Giant California kelp

# OIL YIELDING PLANTS

## ○ U-forbia Letharis

- Carbohydrate → hydrocarbon
- Grow in desert
- Require less water
- Petroleum products can be obtained.

## ○ Kopaiba

- Grown in Brazil.
- Mineral oil obtained from the stems of these plants.
- Holes are drilled on the trunks of the plants, through pipe



# OIL YIELDING PLANTS

## ○ Jojoba

- Grown in Mexico
- It gives seeds which give oil
- Used as fuels for vehicles

## ○ Palm trees

- Grown in coastal areas
- Leaves and stems are cut & dried to obtain fuel.

## ○ Coconut tree

- Grown in coastal areas
- Direct oil is used
- Stems & fibers are used in biomass

# OIL YIELDING PLANTS

- Eucalyptus Trees
- Fast growing trees
- Wood is used as fuel
- Leaves give oil, used for pharmaceutical purpose

# ADVANTAGES OF ENERGY PLANTATION

- Wood is used as domestic fuels
- Biomass energy can be obtained from it
- Keeps atmosphere pollution free
- Maintain ecological balance
- Provides Renewable energy source
- Plantation can be done at any place
- Provide other forest products like fruits,gum,honey
- Natural oil is available, used as fuel as well as lubricant
- Provides raw materials for soap,shampoo,car police,shoe polish,printing ink,packing material

# CHARACTERISTICS OF BIOMASS

- All time valuable and never exhausting
- It doesn't contribute in pollution & global warming
- It have high calorific value and can be used for power generation.
- Raw energy freely available everywhere
- Conversion technology is simple. Unskilled person can use it.
- Biogas produced from biomass can be burnt directly in gas stove just like LPG stoves.
- Biomass conversion gives good quality organic manures
- It is an organic matter.

# USE OF BIOMASS

- Waste organic biomass can be directly used as domestic fuels.
- Used as domestic fuels in gas stoves like LPG gas
- Used to run engines, boilers and turbines
- Methane gas can be used to run the gas engines and farm machineries.
- Used for heating the water.

## 5.3 TYPES OF BIOMASS CONVERSION PROCESSES

- Thermo chemical processes
  - 1) Pyrolysis-
  - 2) Hydrogasification
  - 3) Hydro generation
- Bio-chemical processes
  - 1) Aerobic treatment
  - 2) Anaerobic treatment
  - 3) Chemical treatment

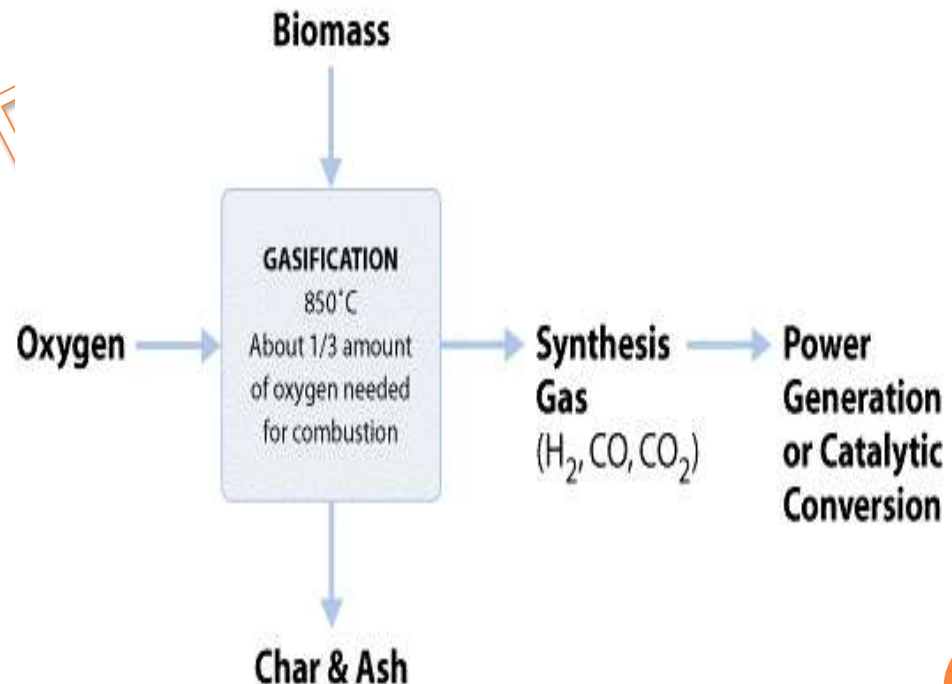
# PRODUCTS WE GET FROM BIOMASS PROCESSES

- Thermo chemical processes
  - 1) Pyrolysis- **Charcoal, Oil, gas**
  - 2) Hydrogasification -**Gas**
  - 3) Hydro generation- **Liquid fuel**
- Bio-chemical processes
  - 1) Aerobic treatment – **Methane, Ethanol**
  - 2) Anaerobic treatment –**Animo Acid,Hydrogen, CO<sub>2</sub>, Methane**
  - 3) Chemical treatment – **Gaseous fuel**

# THERMOCHEMICAL CONVERSION

- Gasification, Pyrolysis, Direct Hydrothermal Liquefaction
- Carbon monoxide and Syngas (Hydrogen)

## Biomass Gasification via Partial Oxidation (Auto Thermal)

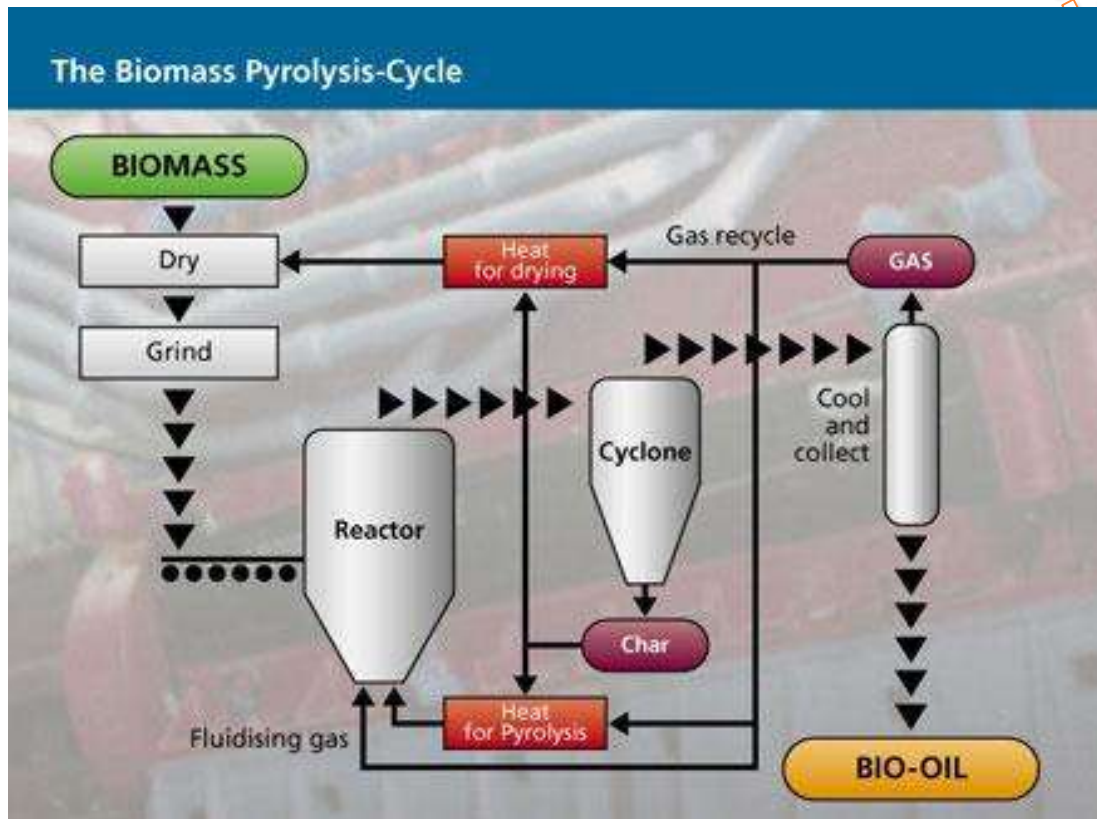


# 1) PYROLYSIS

- Absence of oxygen
- Thermal degradation
- Liquid pyrolysis oil



# PYROLYSIS



# HYDRO GASIFICATION PLANT IN AUSTRALIA



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# ANAEROBIC DIGESTION

- Biogas Platform - Methane
- Decomposition - microorganisms
- Anaerobic Digesters
- Four Main Processes
- Uses wastes and turns into valuable compost



# TRANSESTERIFICATION

- “Biodiesel” Platform
- Takes vegetable oil, animal fat, or grease into biodiesel – fatty acid methyl ester
- Base catalyzed of the oil with alcohol, direct acid catalyzed, and conversion of the oil to fatty acids and then to alkyl esters with acid catalysts



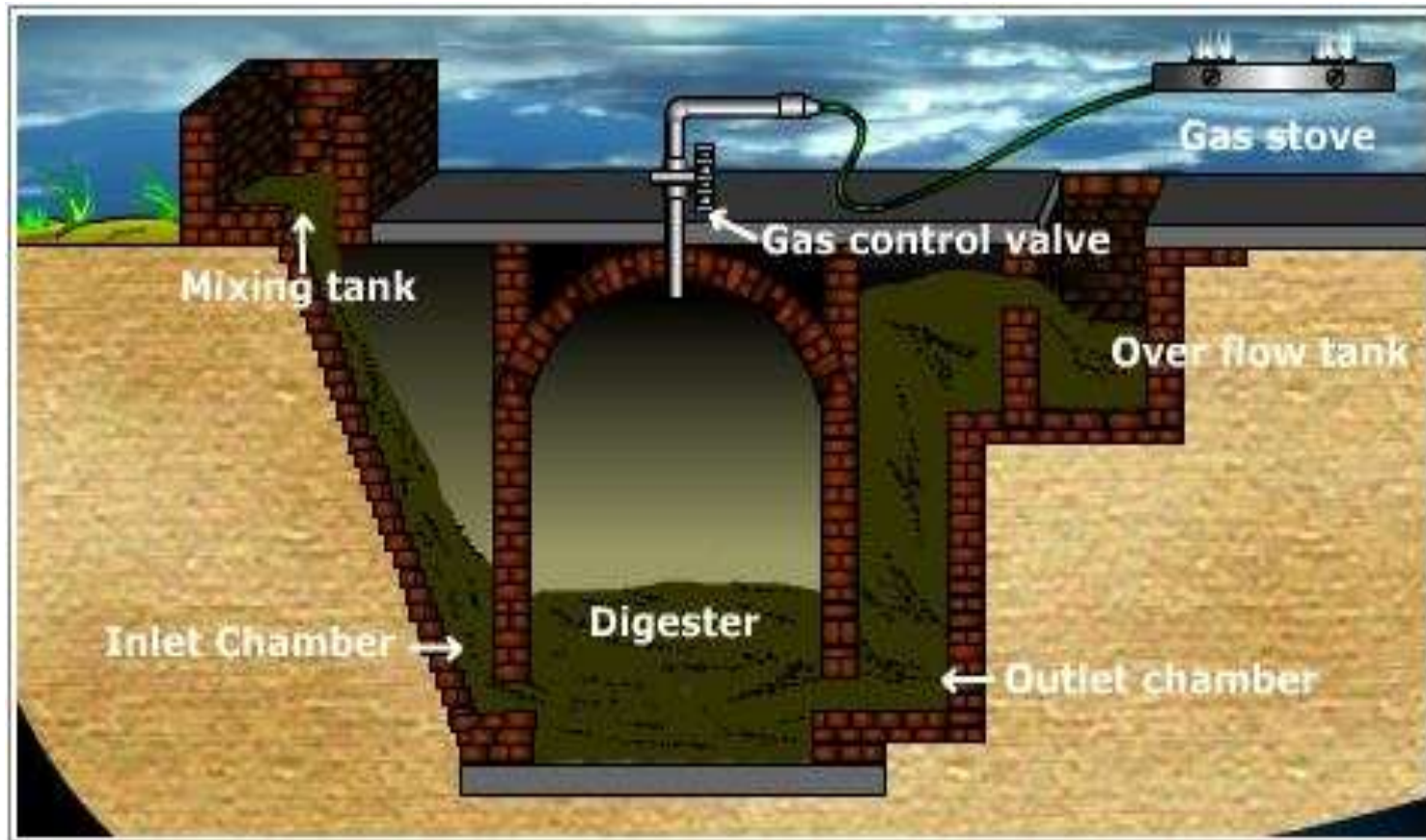
## 5.4 BIOGAS PRODUCTION

- Biogas typically refers to a mixture of different gases produced by the breakdown of organic matter in the absence of oxygen.
- Biogas can be produced by anaerobic digestion with anaerobic organisms, which digest material inside a closed system
- Biogas can be compressed, the same way as natural gas is compressed to CNG, and used to power motor vehicles.

## BIOMASS FEEDSTOCKS

- **Grains and starch crops** – sugar cane, corn, wheat, sugar beets, industrial sweet potatoes, etc.
- **Agricultural residues** – Corn stover, wheat straw, rice straw, orchard prunings, etc.
- **Food waste** – waste produce, food processing waste, etc.
- **Forestry materials** – Logging residues, forest thinnings, etc.
- **Animal byproducts** – Tallow, fish oil, manure, etc.
- **Energy crops** – Switchgrass, miscanthus, hybrid poplar, willow, algae, etc.
- **Urban and suburban wastes** – municipal solid wastes (MSW), lawn wastes, wastewater treatment sludge, urban wood wastes, disaster debris, trap grease, yellow grease, waste cooking oil, etc.

# THE BIOGAS PLANT



# THE MAIN COMPONENTS OF THIS SYSTEM ARE:

- 1) Inlet pipe
- 2) digester tank
- 3) gas holder tank
- 4) slurry outlet pipe
- 5) gas outlet pipe

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# USE OF BIOGAS PLANTS SLURRY

- It is riched in Nitrogen & Phosphorus
- Very fertile and increase the crop production.
- It can be used to obtained nutrients like vitamin B & B<sub>12</sub> in pharmaceutical products.
- Output slurry is used in farms.

# FACTORS AFFECTING THE PRODUCTION OF BIOGAS

- Proportion of biomass and the water in the input slurry
- Rate of feeding the biomass
- Design & construction of biomass plant
- Seeding
- Proportion of Carbon and Nitrogen
- Size of digester
- Average of Temp.
- Pressure of gas
- PH value of biomass
- Nutrients
- Agitation of biomass slurry

THANK YOU...